



Study on how to improve visibility of transparent display for augmented reality under various environment conditions

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Abstract: In this paper, a psychophysical investigation to improve a visibility of a transparent display is presented. A new illuminance measurement method for the transparent display, named eye illuminance, is proposed. Through a psychophysical experiment, it is found that the eye illuminance is strongly related with the visibility of the transparent display regardless of its background condition. This paper finds out the optimum emission luminance range of the transparent display under various illuminant conditions. Also, the contrast ratio for visibility is analyzed and it is found that a higher contrast ratio is not needed to provide a visually better image under a brighter ambient environment. In conclusion, our findings will contribute to an auto brightness control technology to improve the visibility of the transparent display for augmented reality devices.

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1. Introduction

Augmented reality (AR) has been studied for many applications such as commerce, education, games, navigation, and so on [1]. AR provides illusion that virtual objects exist in a real space by showing virtual objects superimposed with the surrounding real environment through an AR display. A transparent display has been mostly used for AR device and various studies have been performed [2–4] to improve its performance. Not only light emitted from the image on the transparent display but also transmitted light from surroundings are perceived simultaneously when a user sees the image on the transparent display. This can be expressed as follows:

$$L_{display} = L_{em} + L_{bg} + L_{gl} \quad (1)$$

where, $L_{display}$, L_{em} , L_{bg} , and L_{gl} denote the total luminance entering the user's eyes, the emission luminance of luminous image on the transparent screen, the luminance of transmitted light through the transparent display from background, and glare luminance reflected by the surface of the display, respectively, as shown in Fig. 1.

The visibility of transparent displays becomes worse by the lights ($L_{bg} + L_{gl}$) from ambient environment than conventional non-transparent counterparts. Hence, various studies have been conducted on how the images on the transparent display are perceived [5–11]. They focused on the visibility influence by features of the transparent display such as transmittance, pixel structure, and ambient condition. Kim *et al.* verified transparent effect on the gray scale perception of a transparent display under ambient conditions [10]. Park *et al.* simulated a visual quality of the transparent display under various ambient light [11]. For the transparent displays, the background light is seen as well as the emitted light from the transparent display. The background light is determined by not only light from light sources but also light reflected by objects. Thus, the background conditions also influence on images of the transparent display [12,13]. Whereas Kim [10] and Park [11] considered only the lights from the light sources, Juan *et al.* [12] and Hincapié-Ramos *et al.* [13] focused on the background conditions. Nevertheless, there have been

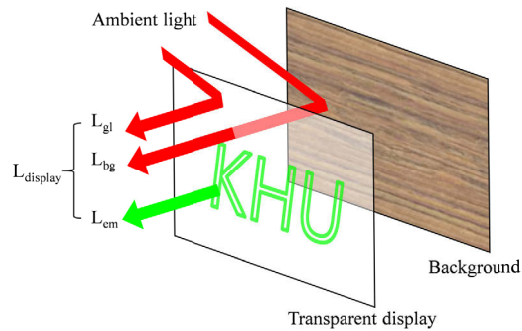


Fig. 1. Three kinds of light from the transparent display.

reported few studies on the display luminance and the visibility of the transparent display under ambient conditions.

The display brightness affects not only the visibility but also a power consumption. For that reason, an auto-brightness control (ABC) technology that adjusts display luminance depending on the ambient illuminance is universally being used in smartphones [14,15]. It helps to improve a quality of experience (QoE) by maintaining the visibility under various environmental conditions. Furthermore, a running time of device can be extended because it reduces waste of energy to operate its display. Most of AR devices are used both indoors and outdoors. Thus, the ABC technology is essential for AR devices. In order to use this technology properly, the optimum display luminance data depending on ambient condition is necessary.

In this paper, we propose an appropriate luminance range and a practical illuminance measurement method that directly affects the visibility of the transparent displays.

The rest of this paper is organized as follows: In section 2, the transparent display used in our experiment and newly defined terms are introduced. The setup and procedure of our experiments in Section 3, experimental results in Section 4, and conclusion in Section 5 will be presented.

2. Transparent display

2.1. System of the transparent display

The transparent display system in our experiment is composed of a laser system and a sheet of quantum dot (QD) film with a transmittance of 82% at 500-800 nm. As shown in Fig. 2(a), the laser system consists of two components: power controller that deals with the intensity of the laser and a laser scanner that handles laser light paths. The QD particles of the film exposed to the light of the laser emit new light with longer wavelength than that of the laser light. By controlling the light path of laser, the laser scanner draws figures or characters on the QD film as shown in Fig. 2(b).

2.2. Two illuminance measurement method

Generally, the ambient illuminance has been measured with an illuminometer facing upwards. However, under the same illuminance condition, the L_{bg} may vary depending on the background environment because the background objects affect the reflected light. For example, the L_{bg} of the transparent display surrounded by the white background will be higher than that of the transparent display surrounded by the black background. Thus, we hypothesized that the illuminance measured near user's eyes facing the display reflects both the illuminance condition and the influence of the background. Thus, we suggest a new illuminance measurement method as shown in Fig. 3 and named as follows:



Fig. 2. (a) A laser system and (b) an image of the transparent display used in the experiment.

- Eye illuminance: illuminance measured by the meter facing the display near user's eyes.
- Ambient illuminance: illuminance measured by the meter facing upwards near the transparent display

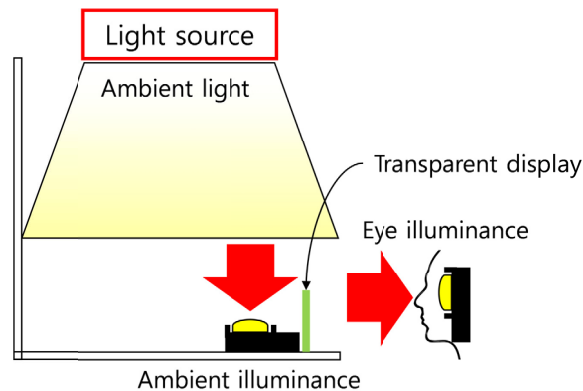


Fig. 3. Two ways to measure illuminance: eye illuminance vs. ambient illuminance.

3. Experiments

3.1. Design

Figures 4(a) and 4(b) show a top view schematic of the experimental setup and actual photo of the setup, respectively. As shown in Fig. 4(a), the QD film (6cm × 6cm) was placed 60 cm away from the subject. All subjects fixed their chin on a chin-rest placed on a table. The laser system was placed outside of a background board and it was 135 cm away from QD film at a 50-degree angle. The background board surrounding the QD film was used for various background conditions. The background conditions were changed by covering the background board with colored paper. A controllable lighting system with high-power light emitting diode (LED) lamps was used and illuminance conditions were also changed by controlling its power.

3.2. Conditions

The experiment consisted of three sessions depending on the colors (white, gray, and black) of the paper covering the background board. In the experiment, the ambient light and emission luminance of transparent display were controlled. We measured the illuminance using an illuminance meter

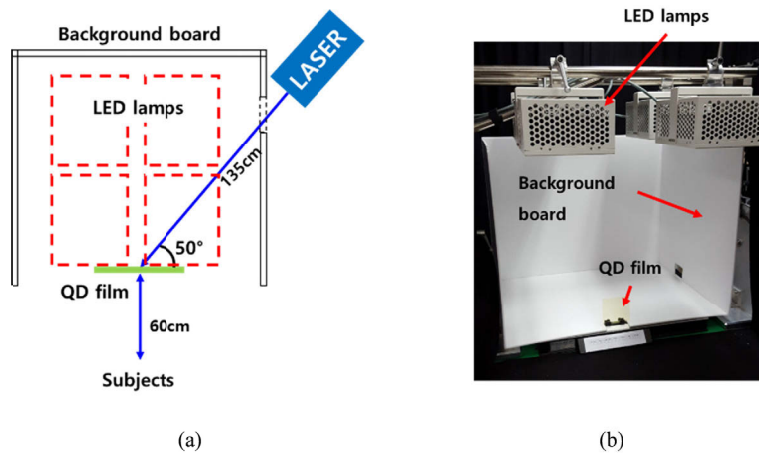


Fig. 4. (a) A top view and (b) an actual photo of the experimental setup.

(Konica Minolta, T-10A) and specified the eight levels of illuminance condition as described in Table 1. Figure 5 represents actual photos of the setup with three different background boards under the ambient illuminance of 2,000 and 5,000 lx. All photos were taken with an aperture of $f/4$ and an exposure time of 1/60 second. As shown in Fig. 5, the ambient brightness depends on the background board as well as the ambient illuminance. The eye illuminance per ambient illuminance level was measured depending on background condition. The eye illuminance levels depending on the background conditions were all different as described in Table 1. The ambient illuminance levels changed from 100 to 30,000 lx sequentially through all three sessions.

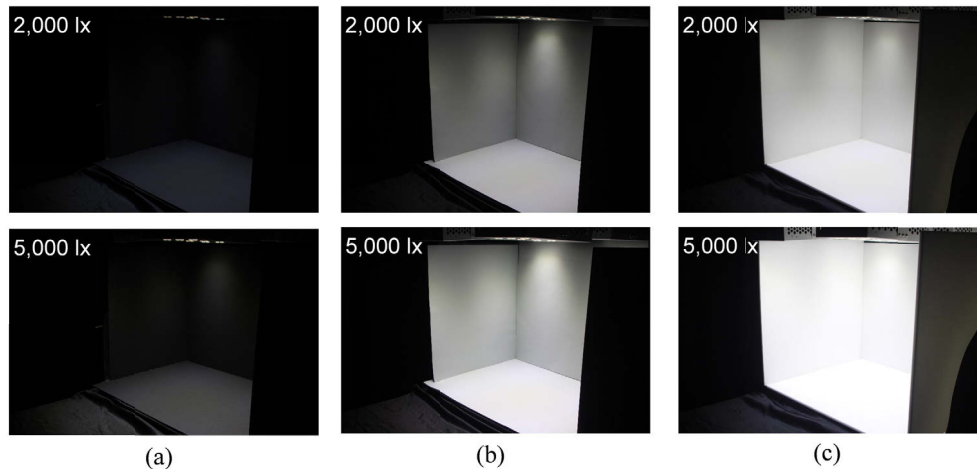


Fig. 5. The photos of the setup with (a) black, (b) gray, and (c) white background conditions under the ambient illuminance conditions of 2,000 and 5,000 lx.

Four levels of L_{em} (26, 61, 95, and, 138 cd/m^2) were used. The L_{em} was randomly selected among four levels and repeated four times in each ambient illuminance condition.

Totally thirty college students participated in the experiment, ten participants per each session. Each subject participated in only one session. The average age of the subjects was 25.3 years.

Table 1. Ambient and eye illuminance levels.

Ambient illuminance [lx]	Eye illuminance depending on background color [lx]		
	White	Gray	Black
100	17	10	3
500	82	48	14
1,000	169	97	28
2,000	346	198	55
5,000	851	483	137
10,000	1,680	970	273
20,000	3,490	2,029	571
30,000	5,040	3,042	886

3.3. Procedure

An image of certain L_{em} was displayed to the subjects as shown in Fig. 2(b). We asked the subjects to respond to experimental stimuli in seven-point Likert scales as shown in Table 2. After the subject answered, the L_{em} was changed randomly among four levels. This procedure was repeated 16 times in each ambient illuminance level. After the 16 repetitions under a certain illumination condition, the ambient illuminance was changed to the next level. And the same procedure was repeated. Finally, the subjects answered 128 times in a session and one session took about 15-20 minutes.

Table 2. The perceived brightness scores

Too dark	Dark	Little dark	Appropriate	Little bright	Bright	Too bright
1	2	3	4	5	6	7

4. Results

4.1. Proposed illuminance measurement method

Figures 6(a) and 6(b) show that the average perceived brightness scores depending on ambient and eye illuminance, respectively. The black squares, gray triangles, and white circles represent background conditions of black, gray, and white, respectively.

We analyzed the experimental results depending on ambient illuminance and eye illuminance. Figure 6(a) shows that the average perceived brightness scores depending on ambient illuminance levels. The average perceived brightness scores decreased as the illuminance increased for the same emission luminance. And they increased as the L_{em} increased. The two observations were the same regardless of types of illuminance (ambient or eye illuminance). However, distributions of data were different depending on illuminance type. The average perceived brightness scores spread depending on the background conditions under ambient illuminance even though both L_{em} and ambient illuminance were the same as shown in Fig. 6(a). For example, in the graph of L_{em} of 138 cd/m^2 , results for black background condition always were higher than those for other background conditions. And the results for gray background condition were higher than those for white counterpart. The values of R^2 , coefficient of determination, were 0.74, 0.74, 0.76, and 0.72 for L_{em} of 26, 61, 95, and 138 cd/m^2 , respectively for ambient illuminance. On the other hand, the values of R^2 were larger than 0.9 for the eye illuminance as shown in Fig. 6(b). From these results, we can conclude that the eye illuminance is closely related with the visibility of the transparent display rather than the ambient illuminance.

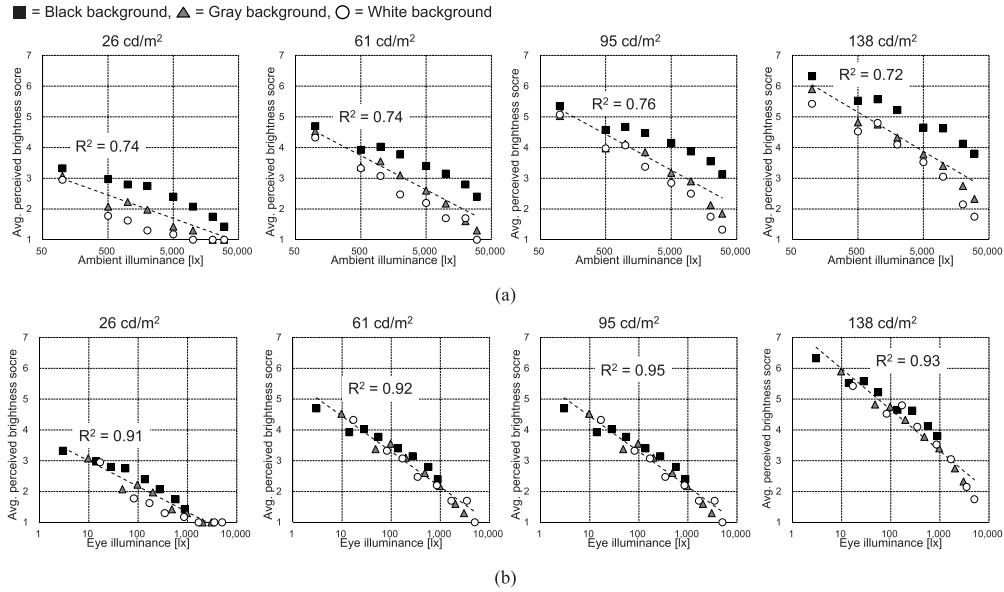


Fig. 6. The experimental results of average perceived brightness scores depending on (a) ambient and (b) eye illuminance.

The significant factors for the eye illuminance are the position where the illuminometer measures and the direction where it faces. A key point of the eye illuminance is to detect intensity of light entering the eyes. From this, an adaptation condition of the eyes determining the visibility can be obtained. Therefore, the eye illuminance is more helpful to determine a reasonable display luminance for higher visibility. It can be simply utilized in the AR devices by placing illuminance sensor in the direction perpendicular to the screen as shown in Fig. 7.

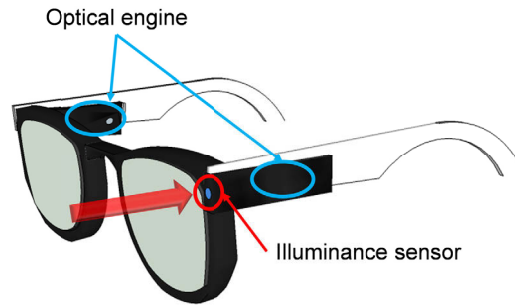


Fig. 7. Position and direction of the illuminance sensor in the AR device to measure the eye illuminance.

4.2. Appropriate emission luminance

Under each eye illuminance condition, the perceived brightness scores for four L_{em} conditions were obtained. From the results, an appropriate L_{em} (L_{appr}) that corresponds to the brightness score 4 depending on eye illuminance levels was derived by using inter-/extrapolation. Figure 8(a) shows that L_{appr} that user feels that a display brightness is appropriate depending on the eye illuminance. Through the regression analysis using SPSS software, we derived the equation which represents the relation between the L_{appr} and the eye illuminance. Their relation can be

expressed as the following equation with $R^2 = 0.96$:

$$L_{appr} = 6.9 \times (ILL_{eye})^{0.49} \quad (2)$$

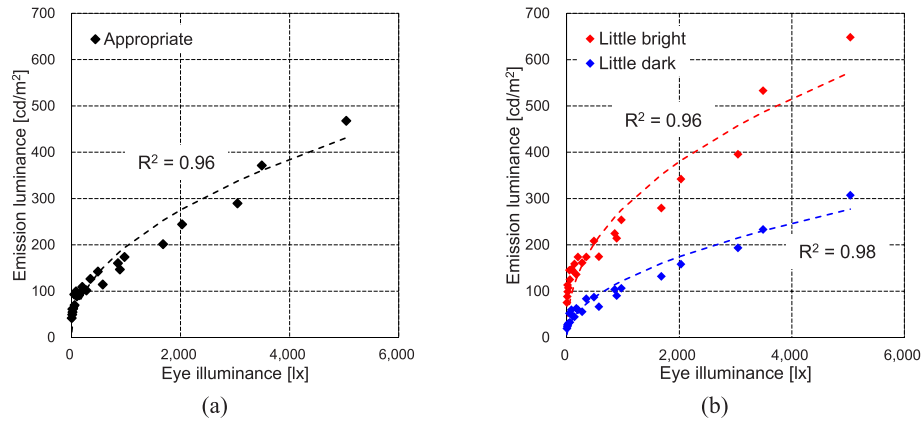


Fig. 8. The emission luminance (L_{em}) that user feels that a display brightness is (a) appropriate, (b) little bright, and little dark depending on the eye illuminance levels.

In Eq. (2), ILL_{eye} denotes the eye illuminance. From Eq. (2), the L_{appr} level under any illuminance conditions can be obtained easily. However, everybody has different preference for the display brightness. For example, the L_{appr} may not be suitable for some users who prefer a brighter or darker display brightness. Thus, we additionally derived two L_{em} levels corresponding to the perceived brightness score 3 and 5, respectively. Figure 8(b) shows that a little bright (red) and a little dark (blue) L_{em} levels depending on the eye illuminance levels. In the graph, symbols denote the obtained values using inter-/extrapolation. And dashed lines denote their trend lines with $R^2 = 0.96$ and 0.98 .

As a results, the L_{appr} can be expressed as a range between the little dark and the little bright L_{em} levels as follows:

$$3.9 \times (ILL_{eye})^{0.50} < L_{appr} < 12.9 \times (ILL_{eye})^{0.45} \quad (3)$$

In Eq. (3), equations of left and right sides are the little dark and the little bright L_{em} levels, respectively. If the transparent display has a L_{em} below the little dark L_{em} level, people may feel that the display is too dark to watch. In order to prevent an unbearably bright transparent display, its L_{em} should be lower than the little bright L_{em} level. For example, under eye illuminance of 1,000 lx, the L_{em} should be between 123 and 279 cd/m² to maintain good visibility. The optimum screen luminance of the transparent display can be automatically adjusted once we know the eye illuminance.

4.3. Contrast ratio

The contrast ratio (CR) is one of the crucial factors to determine the display visibility. In order to verify if the contrast ratio would be affected by transparency, we analyzed the contrast ratio for visibility of the transparent display. The CR is defined as the ratio of the brightest luminance to the darkest one of a display. However, since the ambient light can transmit partially the transparent display, the luminance of the transmitted light has to be considered. Therefore, when applied to this, the brightest luminance is the sum of emission luminance, background luminance, and glare. From here, let's call the sum of L_{bg} and L_{gl} as a base luminance (L_{base}) of

the transparent display. Then, the darkest luminance is the base luminance. Thus, the contrast ratio of transparent display (CR_T) is expressed as follows:

$$CR_T = \frac{L_{em} + L_{base}}{L_{base}} \quad (4)$$

The L_{base} levels depending on the eye illuminance were measured as shown in Fig. 9. In order to obtain the L_{base} , we measured the surface luminance of the QD film using a luminance color meter (TOPCON, BM-5AC) under illuminance conditions as described in Table 1. During the measurement, the QD film did not emit light and just the illuminance condition was changed. There is a strong linearity between the L_{base} and eye illuminance with $R^2 = 0.98$. Thus, the L_{base} can be expressed as the following equation:

$$L_{base} = \alpha \times ILL_{eye} \quad (5)$$

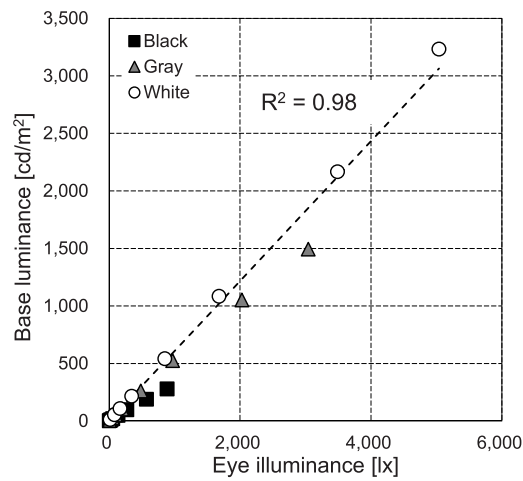


Fig. 9. The base luminance (L_{base}) depending on the eye illuminance levels.

In Eq. (5), α denotes a proportional factor and it was 0.67 in our experiment. A value of α could vary depending on some features of the transparent display, e.g. transmittance or a measurement distance. However, because the L_{base} has a linear relationship with the eye illuminance, the value of α can be simply obtained once we measure L_{base} for a certain eye illuminance condition.

Let's replace L_{base} of Eq. (4) with L_{base} of Eq. (5), then the CR_T could be expressed as follows:

$$CR_T = \frac{L_{em}}{\alpha \times ILL_{eye}} + 1 \quad (6)$$

Equation 6 shows that the CRT can be estimated from the eye illuminance. For example, if it is applied to a device, by measuring the eye illuminance, the L_{em} could be adjusted to maintain the contrast under various environments.

In order to analyze the contrast ratio when the subjects felt the appropriate brightness for the transparent display, we converted the L_{appr} to the CR_T . Figure 10 shows the CR_T values depending on the eye illuminance by using data shown in Fig. 8(a). We expected that the CR_T would be constant within a range regardless of the ambient condition. But contrary to our expectation, it was observed that the CR_T decreased as the eye illuminance increases and was closer to 1 under extremely brighter environment. This is very interesting, and we believe that our new finding deserves to be shared with researchers and engineers in the display field. This result shows that there is no need to aim high level of CR_T for the AR devices under a very bright environment.

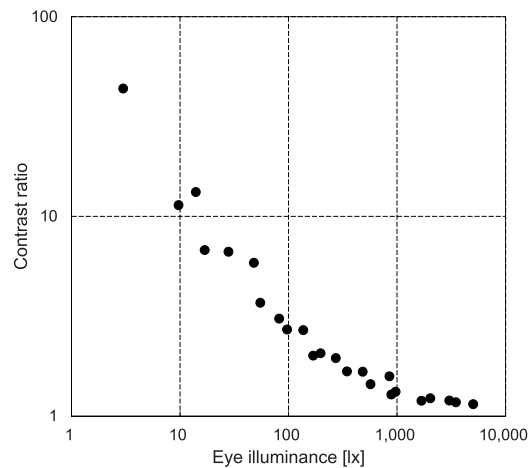


Fig. 10. The contrast ratio depending on the eye illuminance levels.

5. Conclusion

We reported psychophysical research findings to improve the visibility of the transparent display under various ambient environments. First, we suggested the eye illuminance. It is very useful illuminance measurement method to anticipate the visibility of the transparent display. It was verified that the eye illuminance is strongly related to the visibility of the transparent display. Thus, it can be used as an index to estimate visibility condition for AR devices. Furthermore, it would provide a clue to determine the position and direction of the illuminance sensor when designing AR devices. Second, we presented the appropriate luminance range depending on the eye illuminance. We expect this finding could be applied even if the transmittance of the display is different. Third, we analyzed the contrast ratio for the visibility of the transparent display. We found out that the contrast ratio of the transparent display could be calculated from the eye illuminance. In addition, the experimental results show that a higher contrast ratio would not be required for better visibility of the transparent display under a brighter ambient condition. It is expected that this finding could be utilized when the transparent display is applied to a device used outdoors. The results of this study provide a basis for the ABC technology. Thus, the ABC technology could be applied to AR devices more easily. As a result, we believe that our findings will contribute to improvement of the visibility and to power saving of the AR devices under various environments.

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Disclosures

The authors declare that there are no conflicts of interest related to this article.

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